



长安链
ChainMaker

长安链开发大赛作品 技术报告

学 校：北京邮电大学

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作品题目：AEGISSENSE: 基于长安链的 WAF 日志防篡改体系与低
误报智能检测系统

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长安链开发大赛技术报告

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参赛题目	AegisSense:基于长安链的 WAF 日志防篡改体系与低误报智能检测系统						
报告摘要	针对 Web 应用防火墙（WAF）日志在实际应用中面临的易被篡改与高误报率双重挑战，本作品提出并实现了一个融合区块链与机器学习技术的 WAF 日志存证与分析框架。本报告将重点阐述该框架的设计与实现，特别侧重于如何利用长安链（ChainMaker）构建可信、防篡改的日志存证体系。框架采用链上链下协同的混合存储架构，将海量日志存储在链下数据库，同时将其密码学哈希摘要与关键元数据通过智能合约存证于长安链，从而在保障数据完整性与可审计性的前提下，兼顾了系统性能与存储效率。在分析侧，框架通过多层特征处理流程有效降低误报。实验结果表明，基于长安链的存证机制对各类篡改行为的检测准确率达到 99.8%，同时整体框架在极端类别不平衡的数据集上实现了低至 0.0005 的误报率与高达 0.9483 的召回率，显著提升了 WAF 日志管理的安全性与实用性。 团队成员将相关成果以一作身份发表于区块链与信息安全国际会议 IEEE ICBCTIS 。						
关键词	Web 应用防火墙；长安链；区块链存证；日志完整性；防篡改；攻击检测						

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1. 引言

Web 应用防火墙（WAF）作为网络安全防御的核心组件，通过实时监控 HTTP/HTTPS 流量产生海量的安全告警日志。这些日志是进行安全事件回溯、攻击取证和威胁分析的关键依据。然而，当前 WAF 日志管理系统面临两大严峻挑战：其一，传统集中式或分布式日志存储架构缺乏有效的防篡改机制，攻击者或内部恶意人员可能篡改或删除日志以掩盖攻击痕迹，导致事后审计与取证失效，破坏了日志的可信性；其二，基于静态规则的检测方法适应性有限，产生大量误报，严重干扰安全人员对真实威胁的识别与响应。

为解决上述问题，本作品设计并实现了一个名为 BML-WAF 的集成框架。该框架创新性地将长安链区块链技术与机器学习分析能力相结合，构建了一个覆盖日志全生命周期的安全处理管道。本报告将详细阐述该框架的架构设计，并着重说明长安链如何在其中扮演确保日志完整性、提供可验证存证的核心角色，从而为 WAF 日志的可信存储与高效审计奠定坚实基础。

2. 基于长安链的 WAF 日志防篡改存证与可信审计系统设计与实现

2.1 作品概述

本作品的核心目标是构建一个安全、可信、高效的 WAF 日志管理系统。系统设计遵循“存证可信、分析精准”的原则，分为两大核心模块：

（1）基于长安链的可信存证模块：利用长安链的不可篡改、分布式共识特性，为每一条 WAF 日志建立唯一的、可验证的“数字指纹”（哈希值）并上链存证。任何对原始日志的篡改都会导致哈希值变化，从而在审计时被迅速发现并告警。

（2）基于多层特征融合的智能分析模块：对已存证的可信日志进行深度分析，通过数据预处理、特征提取、规则修正与多特征融合等步骤，精准计算每条日志为真实攻击的概率，有效抑制误报，提升威胁检测的准确性。

两大模块协同工作，形成了从日志生成、安全存证到智能分析、结果输出的完整闭环，为 WAF 的运维提供了从“数据可信”到“分析可靠”的全方位保障。

2.3 背景分析

当前，保障日志完整性的方案多依赖于集中式可信第三方或简单的哈希校验，存在单点故障或校验凭证本身易被篡改的风险。区块链技术，以其分布式、不可篡改、可追溯的特性，为日志存证提供了理想解决方案。然而，将海量 WAF 日志直接全部上链会面临区块链吞吐量低、存储成本高昂的挑战。

长安链作为一款国产自主可控的区块链底层平台，其高性能并行共识机制能有效克服日志存证操作中的吞吐量瓶颈，并且其原生支持的可定制化智能合约自动化证据锚定提供了便利。这些特性使其非常适合作为本框架防篡改存储层的底层基础设施。

因此，本作品采用混合链上/链下存储架构：原始日志经处理后存储于高效的链下数据库（可信数据库），同时计算其哈希值并将该哈希与日志元数据（如存储路径、时

间戳)一同构成存证交易,通过智能合约提交至长安链网络。此设计既利用了区块链的防篡改优势,又避免了其存储性能的局限,是兼顾安全与效率的优选方案。

2.4 架构设计/技术路线

本框架整体采用分层架构设计,共包含五个核心层次:数据预处理层、区块链存证层、特征提取层、规则修正层以及多特征融合层。系统流程始于原始 WAF 日志,终于最终攻击概率分数 S_{final} 。整体架构设计如下图所示:

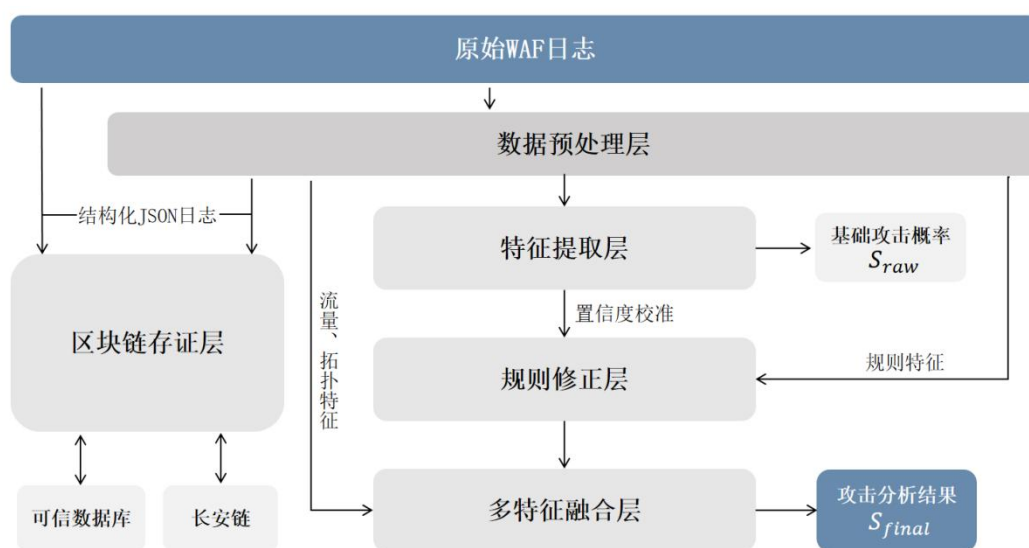


图 1 框架整体设计图

(1) 数据预处理层: 原始 WAF 日志格式异构、结构不一。本层负责将其转换为清洁、统一、适于存证与分析的结构化数据。处理流程包含三个步骤: ①数据清洗: 去除无关条目、消除重复记录、填补缺失值。②字段标准化: 解析并提取关键告警字段(如源 IP、请求 URL、用户代理等), 封装为统一的 JSON 格式。此标准化输出是后续区块链哈希计算和特征提取的基础。③特征增强: 从原始数据中衍生出额外的上下文与拓扑特征(如源 IP 的访问频率), 为下游分析提供更多判别信号。

(2) 区块链存证层: 本层是保障日志完整性与可审计性的核心, 基于长安链构建。它接收预处理层的结构化 JSON 日志, 采用前文所述的混合存储架构。其存证与审计流程清晰分为两个阶段, 如下图所示:

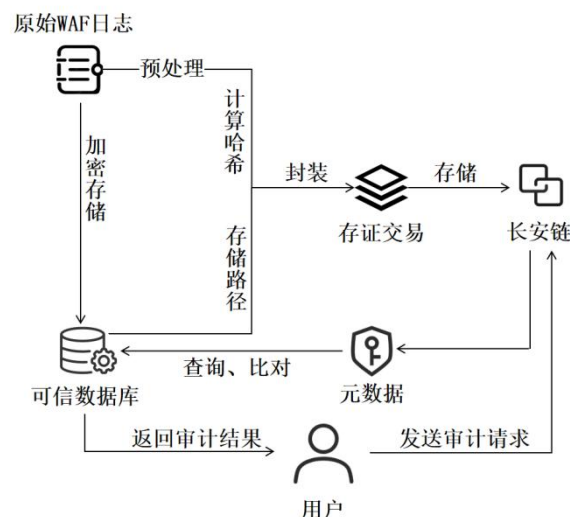


图2 区块链存证层架构图

具体技术实现如下：

存证阶段：①链下存储：将预处理后的结构化JSON日志，加密存储于高性能的链下可信数据库中，并生成唯一访问路径（如URI或索引）。②哈希计算：对同一份JSON日志数据计算其SHA-256哈希值，生成唯一的、不可逆的数字指纹。③链上存证：将计算得到的哈希值、日志存储路径、时间戳等关键元数据，通过调用预先部署在长安链上的智能合约，组装成一笔“存证交易”并提交上链。④共识固化：长安链网络中的共识节点对这笔交易进行验证与共识，一旦达成共识，该存证记录便被永久、不可篡改地记录在长安链的分布式账本上，成为该日志在特定时刻存在且未被更改的权威证明。

审计验证阶段：①凭证查询：用户或审计系统发起验证请求，通过智能合约接口查询长安链上对应日志的存证记录，获取原始哈希值和链下存储路径。②数据获取与重算：根据存储路径，从链下可信数据库中取出待验证的日志文件。③哈希比对：重新计算所取日志的SHA-256哈希值。④结果判定：将新计算的哈希值与从长安链上获取的原始哈希值进行比对。若两者完全一致，则证明日志自存证以来未被篡改，验证通过；若不一致，则表明日志已被篡改，系统立即触发安全告警。

该机制确保了任何对链下数据库中文档的篡改（如内容替换、字段删除、时间戳修改等），都会导致其哈希值与长安链上固化的记录不匹配，从而被高效、可靠地检测出来。

(3) 特征提取层：本层负责将结构化的日志文本转化为机器可读的数值特征。首先，将 JSON 日志中的关键字段（如漏洞类型、攻击载荷）拼接为统一文本序列 T 。随后，使用 TF-IDF（词频-逆文档频率）向量化方法将 T 转换为特征向量 X ，以突出具有区分度的关键词。TF-IDF 的计算基于词频 $TF(t, T)$ 和逆文档频率 $IDF(t)$ ，最终特征值为两者乘积： $TF-IDF(t, T) = TF(t, T) \times IDF(t)$ 。将特征向量 X 输入一个逻辑回归 (LR) 模型，通过线性加权求和 $z = w \cdot x + b$ 及 Sigmoid 函数映射 $\sigma(z) = 1/(1+e^{-z})$ ，得到基于文本语义的基础攻击概率 S_{raw} 。该概率作为后续规则修正的起点。

(4) 规则修正层：基于 S_{raw} ，本层引入领域知识进行校准。通过预定义的安全规则库（如：响应体包含防御关键词、用户代理匹配扫描工具特征、原告警为低风险等级等），对原始分数进行动态权重调整： $S_{adj} = S_{raw} \times w$ （ w 为规则权重系数）。同时，本层会记录触发的规则，并生成一组高可解释性的布尔型规则特征向量 $f_{rule} = [f_{resp}, f_{ua}, f_{low}]$ ，分别对应不同的规则匹配情况。输出为校准后的概率 S_{adj} 和规则特征 f_{rule} 。

(5) 多特征融合层：本层是攻击判定的最终环节，负责融合来自多源的异构特征：
①基础特征 (f_{base})：将 S_{adj} 通过 logit 函数转换，映射到整个实数域。
②规则特征 (f_{rule})：来自规则修正层的布尔向量。
③流量与拓扑特征 ($f_{traffic}$)：从预处理层提取的上下文特征。将融合后的特征向量 $F = [f_{base}, f_{rule}, f_{traffic}]$ 输入最终的逻辑回归模型，计算最终攻击概率 S_{final} ： $S_{final} = \sigma(w^T F + b)$ 。模型训练采用交叉熵损失函数，并为攻击类别设置更高权重以应对类别不平衡。最终，通过优化 F1 分数或满足高召回率（如 $Recall \geq 0.99$ ）下的最高精确率来确定判定阈值。

2.5 模块设计/功能划分

根据架构，系统可划分为以下功能模块：

(1) 日志采集与预处理模块：对接原始 WAF 日志源，完成清洗、标准化和特征增强，输出结构化 JSON 数据。

(2) 长安链存证服务模块：下分为三个子模块，分别为：①哈希计算子模块：负

责对结构化日志计算 SHA-256 哈希。②智能合约交互子模块：封装与长安链智能合约的交互，完成存证交易的构造、发送与状态查询。③链下存储管理子模块：管理可信数据库的读写，维护日志数据与链上存证记录的映射关系。

（3）审计验证服务模块：提供用户接口，受理审计请求，完成从链上查询、链下取数到哈希比对的全流程，并返回验证结果或告警。

（4）特征工程与模型计算模块：下分为四个子模块，分别为：①特征提取子模块：实现 TF-IDF 向量化和基础 LR 模型，产出 S_{raw} 。②规则引擎子模块：加载并执行安全规则库，完成概率校准和规则特征生成。③多特征融合子模块：集成各类特征，运行最终 LR 模型，产出 S_{final} 及可解释信息。

（5）系统管理与配置模块：提供规则库管理、模型参数配置、系统监控等功能。

2.6 接口规范

（1）日志输入接口：接收来自不同 WAF 设备的原始日志流（支持 Syslog、HTTP API 等格式）。

（2）存证交易接口：定义与长安链智能合约交互的 API，包括生成存证交易、查询交易回执等。

（3）审计查询接口：提供 RESTful API，接受包含日志 ID 或时间范围的审计请求，返回存证状态与验证结果。

（4）分析结果输出接口：以结构化格式（如 JSON）输出每条日志的最终攻击概率 S_{final} 、触发规则列表及关键特征贡献度。

2.7 引用其他开源项目及其开源协议

（1）长安链（ChainMaker）：本作品区块链层的底层平台。其开源协议遵循长安链官方发布的相关开源许可证。

（2）机器学习库（如 scikit-learn）：用于实现逻辑回归等模型。遵循 BSD 开源协议。

3. 结论

本作品设计并实现了一个集成长安链存证与机器学习分析的 WAF 日志管理框架。通过构建链上（长安链）链下（可信数据库）协同的混合存储架构，成功解决了日志易篡改的难题，实验验证其防篡改检测准确率高达 99.8%。同时，通过特征提取、规则修正与多特征融合的分析管道，在极端不平衡数据上实现了低误报率与高攻击检出率。本框架的核心优势在于利用长安链的不可篡改特性，为后续所有分析任务奠定了数据可信的基础。未来工作可围绕长安链生态进一步深化，例如：探索利用长安链的隐私计算合约实现跨机构间的安全日志协同分析，或在存证元数据中引入属性基加密策略以实现更细粒度的访问控制，从而在更复杂的多方可信协作场景中发挥更大价值。

相关成果已发表于区块链与信息安全国际会议 IEEE ICBCTIS 2025。

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5. 附录

发表于 IEEE ICBCTIS 2025 的论文全文内容如下（已模糊作者信息）：

BML-WAF: A Framework Based on Blockchain and Machine Learning for Web Application Firewall Log Integrity and Analysis



Abstract—To address the dual challenges of susceptibility to tampering and high false positive rates in Web Application Firewall (WAF) logs, this paper proposes BML-WAF(Blockchain Machine Learning-based WAF), a novel framework that integrates blockchain and machine learning. For secure storage, a hybrid on-chain/off-chain architecture is introduced to guarantee log integrity by immutably anchoring cryptographic hashes on the blockchain. For accurate analysis, the framework employs a multi-stage pipeline featuring semantic extraction, rule-based calibration, and heterogeneous feature fusion to precisely discern real attacks. Extensive experiments on a highly imbalanced real-world dataset demonstrate its exceptional performance: the framework achieves 99.8% tamper detection accuracy and reduces the false positive rate by 77.27% compared to a Random Forest baseline, while also exhibiting superior precision and recall—validating its strong practicality for real-world deployment.

Keywords—Web Application Firewall, Blockchain, Machine Learning, Log Integrity, False Positive Reduction, Attack Detection

I. INTRODUCTION

As a critical component of web security defense systems, Web Application Firewall (WAF) generates massive volumes of security alert logs through real-time monitoring of HTTP/HTTPS traffic. However, existing logging systems face two major challenges: traditional centralized storage architectures lack robust anti-tampering mechanisms, making them susceptible to tampering by attackers who can erase their traces and complicate forensic evidence collection [1]; meanwhile, detection methods relying on static rules exhibit limited adaptability, resulting in persistently high false positive rates [2]. The prevalence of such false alarms significantly

impedes accurate identification and timely response to genuine threats. Consequently, ensuring log integrity, mitigating false alarms, and enabling in-depth analysis have become critical and ongoing research challenges in web security [3].

II. RELATED WORK

In the domain of WAF log storage, both centralized systems like Syslog [4] and distributed systems like Flume [5] fundamentally lack built-in tamper-proofing mechanisms. This inherent vulnerability undermines the evidential value of logs for post-incident forensic analysis. While blockchain technology inherently ensures data integrity, its direct application to voluminous WAF logs is hampered by performance and scalability constraints. Pioneering works by Taguchi et al. [6] and Pourmajidi et al. [7] effectively addressed this by proposing hybrid on-chain/off-chain models, where only cryptographic digests are stored on the blockchain. These approaches provide crucial references for the evidence preservation mechanism in this framework.

To implement the hybrid storage architecture, BML-WAF employs ChainMaker [8] as its underlying blockchain infrastructure. The platform's high-performance parallel consensus mechanism effectively overcomes throughput limitations in log storage operations, while its native support for customizable smart contracts facilitates automated evidence anchoring. These technical capabilities make ChainMaker particularly well-suited for implementing the tamper-evident storage layer, establishing a secure and efficient foundation for subsequent analytical components in the framework.

For WAF log analysis, reliance on static, rule-based methods consistently results in high false positive rates, overwhelming

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security analysts [9]. While machine learning (ML) offers potential for improved accuracy, purely data-driven models often face their own challenges: they may require extensive, manual feature engineering and typically operate as “black boxes,” offering limited interpretability—a significant drawback in security contexts where justifying a decision is crucial [10]. Recent studies have begun to explore pathways to effectively integrate rules with ML. For instance, Arul et al. [9]. combined logistic regression with expert rules, offering valuable insights for the multi-layer feature fusion architecture introduced in this paper.

To systematically tackle the dual challenges of **secure storage** and **accurate analysis** of **WAF logs**, this paper proposes a unified framework, BML-WAF, which incorporates both on-chain/off-chain collaborative storage and multi-layer feature fusion analysis.

III. FRAMEWORK DESIGN

BML-WAF(Blockchain–Machine Learning–based WAF) is designed with a layered architecture for securing and analyzing WAF logs. The overall system, depicted in Fig. 1, is composed of five primary components: the Data Preprocessing Layer, the Blockchain-based Storage Layer, the Feature Extraction Layer, the Rule-based Calibration Layer, and the Heterogeneous Feature Fusion Layer.

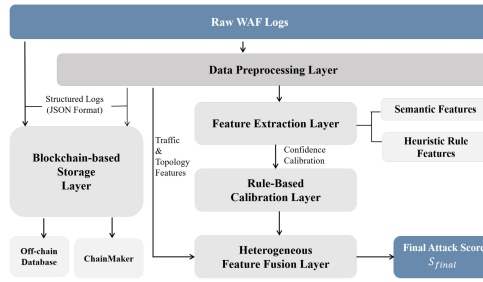


Fig. 1: Framework Overall Design

The BML-WAF framework follows a structured pipeline to ensure both the integrity and intelligibility of WAF logs. The process begins with **Raw WAF Logs** entering the **Data Preprocessing Layer**, executing the essential transformation of this raw data into a cleansed, structured and analysis-ready format. The processed logs then flow through two parallel paths: the **Feature Extraction Layer** converts log content into numerical semantic features, while the **Blockchain-Based Storage Layer** anchors log hashes immutably on the Chain-Maker blockchain, with full logs stored securely in an **Off-chain Database**. The extracted features are subsequently refined by the **Rule-based Calibration Layer**, which incorporates Heuristic Rules to adjust the preliminary attack score. Finally, the **Heterogeneous Feature Fusion Layer** integrates semantic, rule-based, and other engineered features to produce

a **Final Attack Score**, enabling accurate and interpretable threat detection.

A. Data Preprocessing Layer

Raw WAF logs are highly heterogeneous in format and structure [11]. This layer is responsible for transforming raw, unstructured WAF logs into a clean, structured, and analysis-ready format. It involves three sequential steps:

1) *Data Cleansing*: The raw logs are first cleansed by removing irrelevant entries, eliminating duplicates, and imputing missing values to ensure data quality.

2) *Field Standardization*: Key alert fields (e.g., source IP, request URL, user agent) are parsed and extracted into a unified JSON structure. This standardization ensures consistency for subsequent feature extraction and blockchain hashing.

3) *Feature Enhancement*: Additional contextual and topological features are engineered from the raw data. These include traffic frequency from source IPs and categorical encodings for low-risk alert types, providing valuable discriminative signals for the downstream fusion layer. The output of this layer is a standardized log entry, primed for both secure storage and in-depth analysis.

B. Blockchain-based Storage Layer

To establish an anti-tamper log storage system, BML-WAF implements a blockchain-based storage layer leveraging the ChainMaker [8]. This layer accepts structured data from the Data Preprocessing Layer and operates on a hybrid on-chain/off-chain storage architecture, as illustrated in Fig. 2.

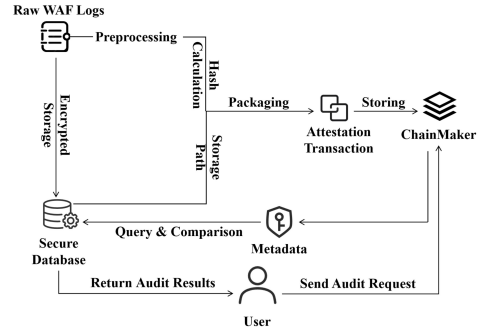


Fig. 2: Blockchain-Based Storage Layer Architecture

During the storage phase, processed logs are stored in the off-chain database, generating a corresponding access identifier. The SHA-256 hash algorithm is applied to compute a unique digest of the log data. This cryptographic hash, combined with relevant metadata, forms an evidence transaction that is submitted through a smart contract [8] to the Chain-Maker blockchain. Following successful consensus validation, the transaction is immutably recorded on the distributed ledger.

During the auditing phase, the on-chain metadata is queried to retrieve the location of the corresponding off-chain log.

Integrity verification is performed by recomputing the hash value of the stored log and comparing it with the immutable record on the blockchain; any discrepancy immediately triggers a security alert. This mechanism not only ensures robust log integrity protection but also provides verifiable traceability for security auditing and compliance requirements.

C. Feature Extraction Layer

This layer converts preprocessed, structured logs into numerical feature representations for attack detection models, emphasizing textual semantic feature extraction and generating a baseline attack probability through logistic regression.

For textual semantic features, key fields (e.g., vulnerability type, attack type) in the standardized JSON are concatenated into a unified text sequence, as shown in (1), where f_n denotes the text content of the i -th field.

$$T = \text{concat}(f_1, f_2, \dots, f_n) \quad (1)$$

Subsequently, the unified text sequence T is mapped to a numerical feature representation by a TF-IDF vectorizer [10], emphasizing discriminative keywords for subsequent modeling. In this process, the term frequency (TF) of a term t in text T is defined in (2), where $\text{count}(t, T)$ denotes the number of occurrences of term t in text T .

$$\text{TF}(t, T) = \frac{\text{count}(t, T)}{\sum_{t' \in T} \text{count}(t', T)} \quad (2)$$

Inverse Document Frequency (IDF) is used to evaluate the global discriminative power of a term. It is calculated as shown in (3), where N is the total number of documents, N_t is the number of documents containing term t , and the addition of 1 is introduced to prevent the denominator from becoming zero.

$$\text{IDF}(t) = \ln \frac{N + 1}{N_t + 1} \quad (3)$$

The value of the TF-IDF characteristic is the product of TF and IDF, as shown in (4).

$$\text{TF-IDF}(t, T) = \text{TF}(t, T) \times \text{IDF}(t) \quad (4)$$

Input the extracted TF-IDF feature vector $X = (x_1, x_2, \dots, x_n)$ into the logistic regression(LR) model [12]. This model calculates the baseline attack probability S_{raw} through a linear weighted sum and a nonlinear mapping.

The LR model assigns weights w_i to each feature x_i and calculates the linear output z together with the bias term b , as shown in (5).

$$z = \mathbf{w} \cdot \mathbf{x} + b = \sum_{i=1}^n w_i x_i + b \quad (5)$$

In (6), the scalar z is mapped to the interval (0,1) via the sigmoid function $\sigma(z)$, yielding S_{raw} .

$$S_{raw} = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (6)$$

The probability S_{raw} provides an initial assessment of attack behavior based on semantic features, which serves as the basis for subsequent rule refinement and multi-feature fusion.

D. Rule-Based Calibration Layer

Based on the attack probability S_{raw} from the feature extraction layer, this module performs intelligent calibration using a predefined security rule library. Dynamically adjusts the raw score to produce a refined score S_{adj} while generating interpretable rule-based features for subsequent multi-feature fusion.

The core score correction applies a weight adjustment strategy, as defined in (7), where w represents an expert-defined rule weight coefficient.

$$S_{adj} = S_{raw} \times w \quad (7)$$

Rule triggers [13] cover multiple security dimensions, including:

- Presence of defensive keywords in response content.
- User agent matching known scanning tool signatures.
- Original alerts classified as low-risk.
- Repetitive behavioral patterns from network entities within time windows.

Upon rule triggering, the weight adjustment mechanism is activated to suppress false positives and filter redundant alerts.

In parallel, the system records the semantic tags associated with the triggered rules and generates key Boolean features, including: response body characteristic f_{resp} , user agent characteristic f_{ua} , and risk level characteristic f_{low} .

The rule refinement layer ultimately outputs the calibrated confidence score S_{adj} along with a set of interpretable rule-based features. These outputs collectively serve as inputs to the subsequent multi-feature fusion layer, facilitating both precise and explainable threat detection.

E. Heterogeneous Feature Fusion Layer

As the final stage of the detection pipeline, this layer integrates multi-source features—including the calibrated probability S_{adj} , Boolean rule-based features from the rule refinement layer, and traffic-topology features from the preprocessing layer—to compute a comprehensive attack probability score.

To adapt the baseline probability for linear modeling, S_{adj} is transformed via a logit function, mapping it to the entire real number space and forming the base feature f_{base} , as defined in (8).

$$f_{base} = \begin{cases} \ln \left(\frac{S_{adj}}{1 - S_{adj}} \right), & \text{if use_logit} = \text{True} \\ S_{adj}, & \text{if use_logit} = \text{False} \end{cases} \quad (8)$$

Rule features f_{rule} are represented by feature vectors composed of three key Boolean variables, as defined in (9).

$$f_{rule} = [f_{resp}, f_{ua}, f_{low}] \quad (9)$$

Each component corresponds to response body whitelist matching, scanner user-agent identification, and low-risk/low-confidence determination, respectively. A value of 1 indicates a rule match, while 0 indicates no match.

In (10), the complete input feature vector $\mathbf{F}$ is constructed by combining the base features, rule features, and the traffic and topology features $f_{traffic}$ from the preprocessing layer.

$$\mathbf{F} = [f_{base}, f_{rule}, f_{traffic}] \quad (10)$$

The fusion process leverages a logistic regression model to compute the final attack probability, applying the linear weighting and nonlinear transformation specified in (11):

$$S_{final} = \sigma(\mathbf{w}^T \mathbf{F} + b) \quad (11)$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the Sigmoid function, $\mathbf{w}$ is the weight vector, and b is the bias term.

The model training utilizes the cross-entropy loss as the optimization objective, as shown in (12).

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (12)$$

where y_i denotes the ground-truth label, $\hat{y}_i$ represents the predicted probability, and N is the total number of samples. To mitigate class imbalance, a class weight of 2.0 is assigned to the attack category during training.

The threshold selection mechanism [14] is based on the precision-recall curve. The optimal threshold θ^* is determined according to the following criterion, as shown in (13):

$$\theta^* = \begin{cases} \arg \max_{\theta} P(\theta), & \text{if } R(\theta) \geq 0.99 \\ \arg \max_{\theta} F_1(\theta), & \text{otherwise} \end{cases} \quad (13)$$

where $P(\theta)$ and $R(\theta)$ denote the precision and recall at threshold θ respectively, and $F_1(\theta)$ is the corresponding F_1 -score.

The fusion module outputs the final attack probability score S_{final} , a list of salient features that contribute significantly to the decision, and the labels of triggered rules. This design enhances interpretability by revealing influential features and rule-based reasoning, thereby achieving a unified framework that balances detection accuracy and explanatory transparency.

IV. RESULTS AND DISCUSSION

A. Dataset

The experiment utilized the publicly available cybersecurity alarm dataset [15], which covers various typical attack types such as SQL injection, command execution, information leakage, and scanning activities. The core information of the dataset is summarized in TABLE I.

TABLE I: Core Information of the Dataset

Attribute	Description
Total Samples	65,134
Attack Samples	291 (0.45%)
False Alarm Samples	64,843 (99.55%)
Data Format	Instruction-Input-Output
Input Format	JSON-formatted Alert Logs
Output Labels	"Attack" / "False Alarm"

The dataset exhibits a severe class imbalance, with false positives accounting for 99.55% of the samples and genuine

attacks making up only 0.45%. This distribution accurately reflects real-world network environments, which are dominated by false alarms [1]. Therefore, it provides an effective basis for validating model generalization and robustness in authentic scenarios.

B. Evaluation Metrics

The core metrics and calculation formulas used to evaluate framework performance are shown in TABLE II.

TABLE II: Evaluation Metrics and Formulas

Evaluation Metric	Formula
False Positive Rate (FPR)	$FPR = \frac{FP}{FP+TN}$
Precision	$Precision = \frac{TP}{TP+FP}$
Average Precision (AP)	$AP = \sum_n (R_n - R_{n-1}) P_n$
Recall	$Recall = \frac{TP}{TP+FN}$
F1-Score	$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$

C. Tamper Evidence Verification

To comprehensively evaluate the framework's capability in safeguarding log integrity, the dataset was partitioned into four distinct test cohorts, each designed to simulate a specific type of tampering operation: content replacement, field deletion, timestamp manipulation, and metadata alteration. From the original collection of 65,134 log instances, 1,000 records were randomly sampled per cohort for targeted tampering. The complete verification workflow is depicted in Fig. 3.

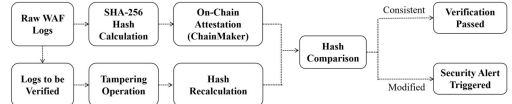


Fig. 3: Tamper-Resistance Verification Workflow

Evaluation results indicate that the blockchain-based hash attestation mechanism exhibits excellent detection performance under various tampering scenarios. As summarized in Fig. 4, the framework achieved an average detection accuracy of **99.8%** over the four defined operation types. Perfect accuracy **100%** was observed for content replacement and field deletion, with timestamp tampering and metadata modification also being identified at high accuracies of **99.7%** and **99.6%**, respectively.

The integrity verification process, in which the system computes the SHA-256 hash of each log entry in real time and compares it precisely against the original hash stored on the blockchain, demonstrated consistently high effectiveness. Although a minimal number of timestamp and metadata tampering cases were not immediately detected due to factors such as system clock synchronization and network latency, all were successfully identified in subsequent batch verification. Overall, any log tampering invariably resulted in a hash mismatch, triggering the security alert mechanism and thereby validating the practical reliability of the proposed framework for protecting WAF log integrity.

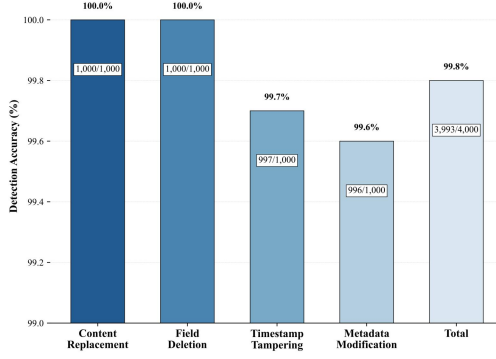


Fig. 4: Detection Accuracy for Different Tampering Types

D. False Positive Reduction

To systematically evaluate the framework's performance in false alarm suppression, three key metrics were adopted: FPR, Precision, and AP. Experimental results demonstrate the framework's outstanding effectiveness in this regard, achieving an FPR of **0.0005**, a Precision of **0.9016**, and an AP of **0.9749**. These results indicate that, even on a highly class-imbalanced dataset, the framework successfully suppresses the vast majority of false alerts while maintaining high discriminative accuracy towards true attack instances.

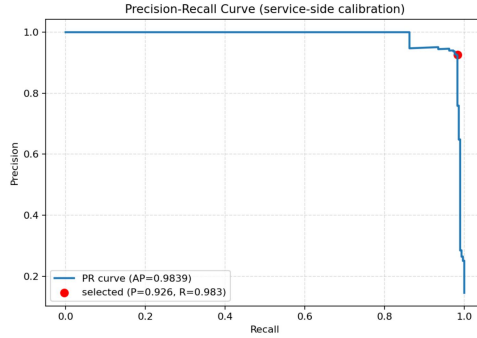


Fig. 5: Precision-Recall Curve

As illustrated in the precision-recall curve of Fig. 5, following server-side calibration, the model achieves a precision of **0.926** and a recall of **0.983** at the operating threshold, with the average precision reaching **0.9839**. This outcome demonstrates that the framework maintains high decision confidence while meeting stringent recall requirements, thereby effectively balancing the critical trade-off between false alarm suppression and attack detection.

E. Attack Detection Performance

The evaluation of attack detection performance focuses on the model's capability to identify genuine threats, with Recall, F1-Score, and AP selected as the core metrics. Experimental results show that the proposed framework achieves a recall of **0.9483** and an F1-Score of **0.9244** on the test set, demonstrating its ability to effectively capture the vast majority of attack activities while maintaining a strong balance between precision and recall. Furthermore, an AP score of **0.9749** confirms the model's robustness in high-precision regions.

As shown in the precision-recall curve in Figure 5, the overall curve lies close to the top-right corner, indicating that the model sustains high precision across varying recall levels. This result validates the framework's capacity to simultaneously achieve high attack detection rates and low false alarms in real-world scenarios.

F. Comprehensive Performance Evaluation

The proposed framework's overall performance for WAF log analysis was evaluated through systematic comparison with Random Forest, a widely adopted and robust baseline. Random Forest [16] was selected based on its demonstrated effectiveness in security detection tasks, capability to handle high-dimensional features [17] while capturing nonlinear relationships, and inherent interpretability [18] through feature importance assessment. The comparative evaluation was conducted under identical experimental conditions, with results detailed in Fig. 6.

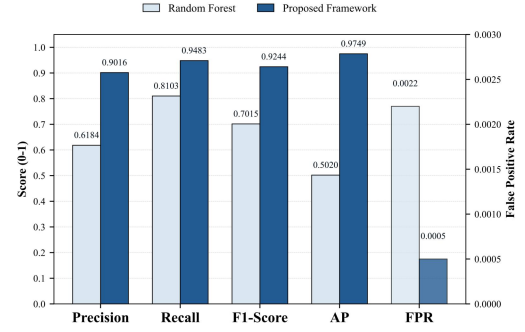


Fig. 6: Key Metrics Comparison

1) False Alarm Suppression: The proposed framework achieves a substantially lower false positive rate of 0.0005 compared to the Random Forest baseline of 0.0022, representing a **77.27%** reduction. This improvement stems from the rule-based correction layer, which incorporates explicit domain knowledge to filter false alarm patterns at the source—a capability lacking in the purely data-driven Random Forest approach. While Random Forest relies solely on statistical correlations learned from training data, the proposed framework

actively eliminates known benign patterns through security-specific rules, thus demonstrating superior performance in minimizing erroneous alerts.

2) *Detection Accuracy and Robustness*: The proposed framework demonstrates superior detection performance with a precision of 0.9016 and F1-score of 0.9244, exceeding the Random Forest baseline by **3.49%** and **2.51%** respectively. This improvement is attributed to the multi-layer feature fusion architecture that systematically integrates complementary feature types, providing more discriminative capability than Random Forest's implicit feature interactions. The framework's explicit feature design enables more reliable decision boundaries, particularly under the challenging class-imbalanced conditions characteristic of security log data.

3) *Attack Detection Capability*: The proposed framework maintains a high recall rate of 0.9483 while achieving an exceptionally low false positive rate, outperforming the Random Forest baseline by **1.41%** in recall. This demonstrates the framework's ability to effectively identify genuine threats without compromising on false alarm suppression—a critical balance that Random Forest fails to achieve due to its inherent limitation in incorporating domain-specific knowledge for precise pattern discrimination. The results confirm the framework's practical advantage in real-world deployment scenarios where both high detection coverage and operational reliability are essential.

In summary, the integration of semantic analysis, probabilistic modeling, rule-based correction, and multi-feature fusion within the proposed framework demonstrates its superior performance over Random Forest across all critical metrics for WAF log analysis.

V. CONCLUSION

This paper proposes BML-WAF, an integrated framework for WAF log management that synergizes blockchain-based integrity protection with machine learning-based analysis. The framework establishes a hybrid on-chain/off-chain storage architecture to ensure log immutability, coupled with a three-layer analytical structure incorporating feature extraction, rule-based correction, and multi-feature fusion for comprehensive threat detection.

Experimental evaluation demonstrates significant performance improvements over the Random Forest baseline across all key metrics. The blockchain-based verification mechanism achieves **99.8%** tampering detection accuracy. In false alarm suppression, the framework reduces the false positive rate by **77.27%** compared to the baseline. For detection capability, it achieves a **3.49%** improvement in precision, **1.41%** higher recall, and **2.51%** better F1-score, while maintaining a **2.17%** advantage in average precision - all accomplished under extreme class imbalance conditions.

Future research directions include implementing attribute-based encryption for fine-grained access control and developing cross-domain collaboration mechanisms to enable secure log sharing and collaborative analysis among multiple organi-

zations, thereby further enhancing the framework's applicability in complex operational environments.

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